

on: 3 Hours

CS1005-1 – INTRODUCTION TO PYTHON PROGRAMMING
(For AI&DS, AI&ML, CB, CCE, CSE, CV, ISE, R&AI)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

1. Who developed Python Programming Language?
A) Wick van Rossum
B) Rasmus Lerdorf
C) Guido van Rossum
D) Niene Stom
2. All keywords in Python are in _____.
A) Capitalized
B) lower case
C) UPPER CASE
D) None of the mentioned
3. Which of the following is the truncation division operator in Python?
A) |
B) //
C) /
D) %
4. What will be the output of the following Python function?
`len(["hello", 2, 4, 6])`
A) Error
B) 6
C) 4
D) 3
5. What is output of `print(math.pow(3, 2))`?
A) 9.0
B) None
C) 9
D) None of the mentioned
6. In order to store values in terms of key and value we use what core data type.
A) list
B) tuple
C) class
D) dictionary
7. What will be the output of the following Python code?
`for i in range(2.0):
 print(i)`
A) 0.0 1.0
B) 0 1
C) error
D) none of the mentioned
8. Suppose `t = (1, 2, 4, 3)`, which of the following is incorrect?
A) `print(t[3])`
B) `t[3] = 45`
C) `print(max(t))`
D) `print(len(t))`
9. _____ represents an entity in the real world with its identity and behaviour.
A) A method
B) An object
C) A class
D) An operator
10. Fill in the line of the following Python code for calculating the factorial of a number.
`def fact(num):
 if num == 0:
 return 1
 else:
 return _____`
A) `num*fact(num-1)`
B) `(num-1)*(num-2)`
C) `num*(num-1)`
D) `fact(num)*fact(num-1)`

11. What will be the output of the following Python code?

```
def fun(n):
    if (n > 100):
        return n - 5
    return fun(fun(n+11));
print(fun(45))
```

A) 50

B) 100

C) 74

D) Infinite loop

12. To open a file c:\scores.txt for appending data, we use _____

A) outfile = open("c:\scores.txt", "a")

B) outfile = open("c:\scores.txt", "rw")

C) outfile = open(file = "c:\scores.txt", "w")

D) outfile = open(file = "c:\scores.txt", "w")

13. What arithmetic operators cannot be used with strings?

A) +

B) -

C) *

D) All of the mentioned

14. Is Python case sensitive when dealing with identifiers?

A) yes

B) no

C) machine dependent

D) none of the mentioned

15. Which of the following statements are true?

A) When you open a file for reading, if the file does not exist, an error occurs

B) When you open a file for writing, if the file does not exist, a new file is created

C) When you open a file for writing, if the file exists, the existing file is overwritten with the new file

D) All of the mentioned

16. The readlines() method returns _____

A) str

B) a list of lines

C) a list of single characters

D) a list of integers

17. How many keyword arguments can be passed to a function in a single function call?

A) zero

B) one

C) zero or more

D) one or more

18. What will be the output of the following Python code?

```
def foo():
    return total + 1
total = 0
print(foo())
```

A) 0

B) 1

C) Error

D) none of the mentioned

19. What will be the output of the following Python code?

```
m = [[x, x + 1, x + 2] for x in range(0, 3)]
```

A) [[1, 2, 3], [4, 5, 6], [7, 8, 9]]

B) [[0, 1, 2], [1, 2, 3], [2, 3, 4]]

C) [1, 2, 3, 4, 5, 6, 7, 8, 9]

D) [0, 1, 2, 1, 2, 3, 2, 3, 4]

20. What is the result of cmp(3, 1)?

A) 0

B) 1

C) True

D) False

PART - B: DESCRIPTIVE ANSWER QUESTIONS.

Unit – I

		Marks	BT*	CO*	PO*
1.	a) What are the various categories or classifications of computers?	07	L*1	1	1
	b) Write a Python Program to display all prime numbers within an interval.	05	L1	1	1
	c) What is the method for adding comments in the Python programming language?	04	L1	1	1
2.	a) Explain block diagram of computer and its components.	06	L1	1	1
	b) Write a Python program to check whether the given integer is a multiple of both 5 and 7.	04	L3	1	1
	c) Write a Python program to find the roots of a quadratic equation.	06	L3	1	2

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|----|----|--|----|----|---|---|
| 3. | a) | What is the sequence of steps involved in program development? | 05 | L2 | 1 | 2 |
| | b) | Provide the algorithm and flowchart for determining the roots of a quadratic equation. | 06 | L1 | 1 | 1 |
| | c) | Write a python program to display all integers within the range 100-200 whose sum of digits is an even number. | 05 | L3 | 1 | 2 |

Unit – II

- | | | | | | | |
|----|----|--|----|----|---|---|
| 4. | a) | What is the difference between a parameter and an argument? | 05 | L2 | 3 | 2 |
| | b) | Write a Python program to find the largest number in a list without using built-in functions | 06 | L3 | 2 | 2 |
| | c) | How can an object be instantiated within a class? | 05 | L1 | 3 | 1 |
| 5. | a) | What are keyword arguments and when should we use them? | 05 | L2 | 3 | 1 |
| | b) | Write a Python program to find the Nth term in a Fibonacci series using recursion. | 06 | L3 | 3 | 2 |
| | c) | Define a Dictionary and write a Python program to print all the items in a dictionary. | 05 | L1 | 2 | 2 |
| 6. | a) | What is the process of creating a tuple and what benefits does it offer? | 05 | L1 | 2 | 1 |
| | b) | Write a Python program to implement matrix multiplication. | 06 | L3 | 2 | 1 |
| | c) | How to Reverse a list using slicing technique? | 05 | L3 | 2 | 2 |

Unit – III

- | | | | | | | |
|----|----|--|----|----|---|---|
| 7. | a) | Write a python program to read the text input from the user and write the contents to a text file. | 08 | L2 | 3 | 2 |
| | b) | How can Python facilitate string manipulation through its supported methods? | 08 | L2 | 3 | 1 |
| 8. | a) | Write a python program to get the full path of the current working directory. | 06 | L2 | 3 | 1 |
| | b) | Write a python program to create a CSV file, insert the contents (USN, Name, Phone number, marks in a subject) related to five students to the file. Sort the contents based on USN and display the top scorer of the subject. | 10 | L2 | 3 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
